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Subject: Advanced Forecasting

Sem: IV, Roll: 0026

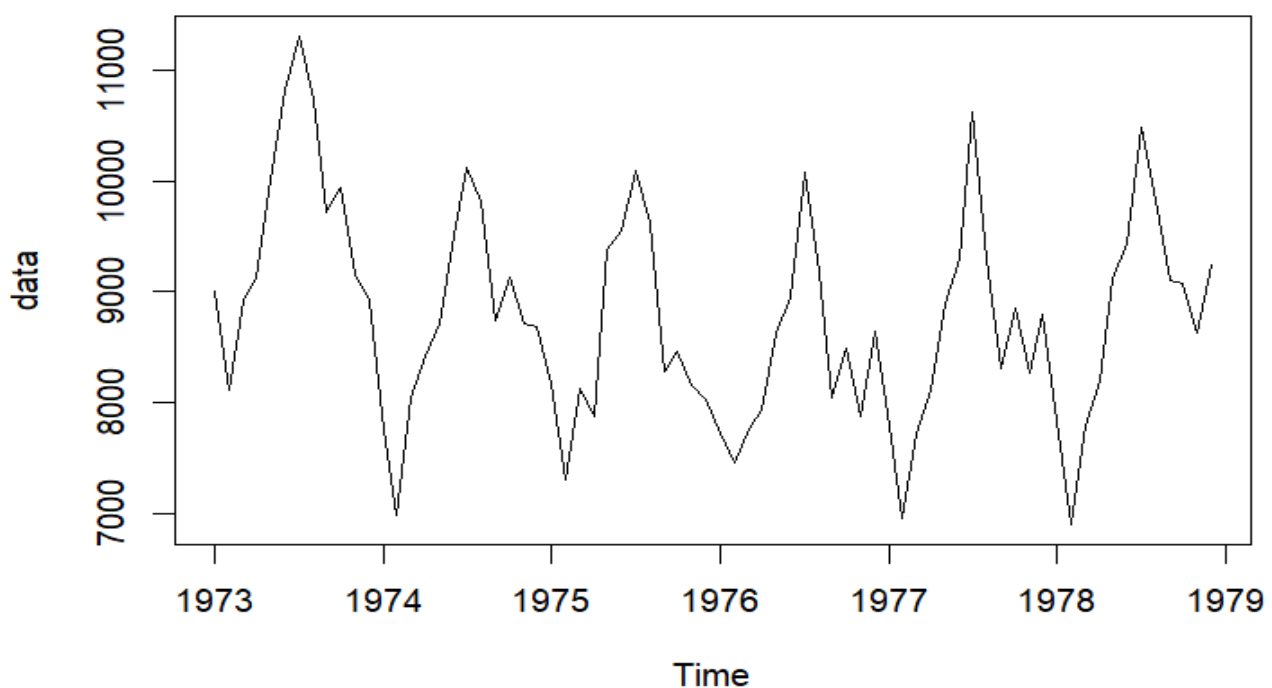
Specialization: Business Analytics

Assignment-I (USAccDeaths)

I have the R Dataset: USAccDeaths (Monthly accidental deaths in the United States).

R Code:

```
library(tseries)
library(forecast)
data=USAccDeaths
View(data)
str(data)
p=as.data.frame(data)
p
plot(p,type="l")
```



Section A: Deep Stationarity Analysis

1. Decompose non-stationarity into trend, seasonal, and variance components.

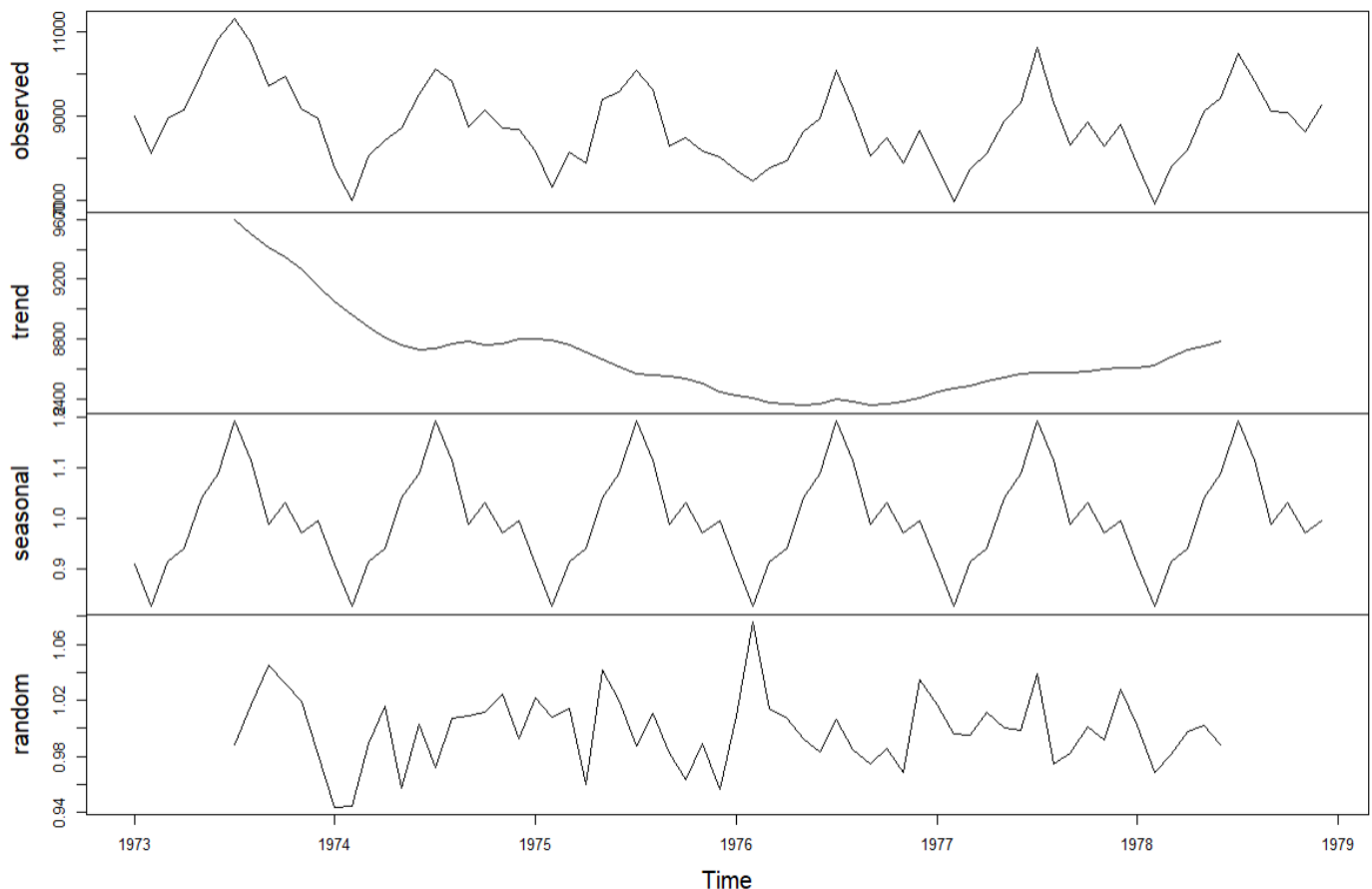
To analyse USAccDeaths (Monthly accidental deaths in the US), we decompose the series X_t into three primary components:

- **Trend Component:** A long-term movement in the data. For USAccDeaths, there is often a slight downward trend over the years.
- **Seasonal Component:** Accidental deaths show strong seasonality, peaking in July (summer travel) and dipping in February (shorter month, less travel).
- **Variance Component:** If the magnitude of seasonal fluctuations increases with the level of the series, a log-transformation is required to stabilize the variance before further analysis.

R Code:

```
ts=ts(p,frequency = 12,start = c(1973,1))
ts
plot(ts,xlab="year",ylab="accidents",main="US Accident Deaths")
library(fpp2)
ts
monthplot(ts,ylab="Accidents",xlab="months")
td=decompose(ts,type="multiplicative")
td
plot(td)
```

Decomposition of multiplicative time series



2. Theoretically justify whether first differencing and seasonal differencing are required.

- **First Differencing:** Required to remove the **stochastic trend**. If the mean changes over time, $(1-B)X_t$ helps achieve a constant mean.
- **Seasonal Differencing:** Required to remove the **seasonal pattern**. Since the data is monthly, $(1-B^{12})X_t$ compares the current month to the same month last year, effectively neutralizing the periodic effect.

3. Over-Differencing and ACF Impact

- **Over-differencing** occurs when a difference operator is applied to a series that is already stationary or has already been appropriately differenced.
- **Impact on ACF:** It introduces artificial dependence. Specifically, the ACF of an over-differenced series will often show a significant **negative spike at lag 1**, and the variance of the series may actually increase rather than decrease.

4. Conflicting ADF and KPSS Results

- **ADF (Augmented Dickey-Fuller):** H_0 is that the series has a unit root (non-stationary).
- **KPSS:** H_0 is that the series is trend-stationary. Conflicting results (e.g., both reject H_0) often occur in the presence of **structural breaks** or when the series has a "near unit root" (high persistence), which weakens the power of the tests to distinguish between a deterministic trend and a stochastic one

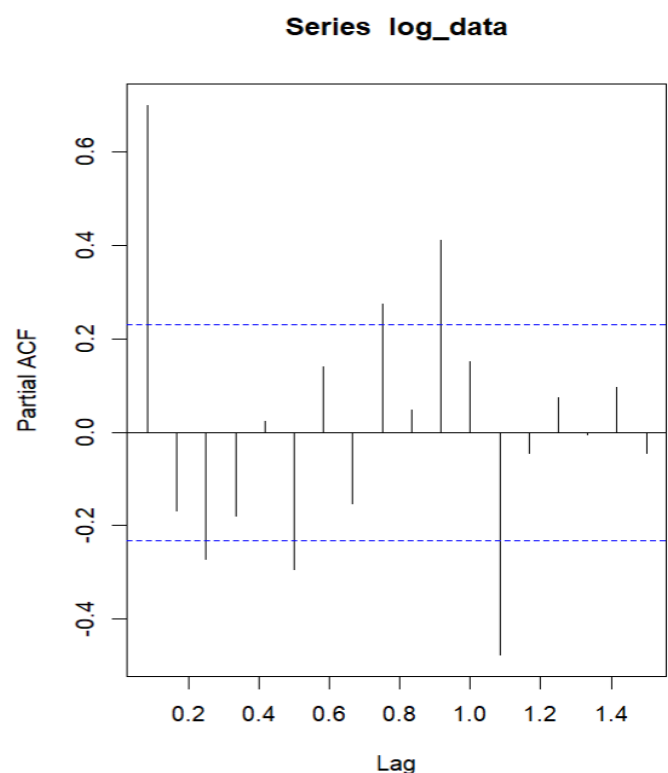
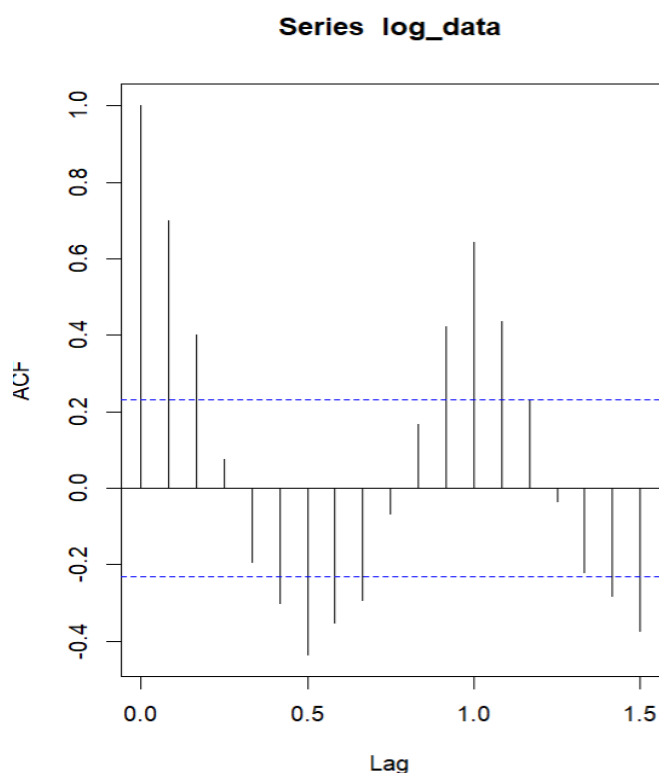
SECTION B: ACF-PACF and Structural Identification

1. Derive expected ACF and PACF patterns for $MA(1) \times SMA(1)$.

The model is expressed as: $X_t = (1 + \theta B)(1 + \Theta B^{12})a_t$.

ACF: Will show significant spikes at **lag 1, lag 11, lag 12, and lag 13**. All other lags will be zero.

PACF: Will show **exponential or sinusoidal decay** at lags 1, 2, 3... and also at seasonal lags 12, 24, 36...



2. Mathematically show why lag 13 appears in ACF.

Expanding the operator $(1 + \theta B)(1 + \Theta B^{12})$ gives:

$$(1 + \theta B + \Theta B^{12} + \theta\Theta B^{13})$$

The interaction term represents the correlation between X_t and the shock at $t-13$, which causes a non-zero ACF value at lag 13.

3. MA vs. AR Behavior

- MA Finite Cutoff: A Moving Average process of order q is a finite linear combination of independent white noise shocks. Once the lag $k > q$, there is no shared shock, so the covariance (and ACF) becomes zero.
- AR Exponential Decay: An Auto-Regressive process is recursive. Each value depends on its predecessor, creating a "chain" of dependence that propagates indefinitely, though the influence diminishes geometrically (exponentially) as the lag increases.

R Code:

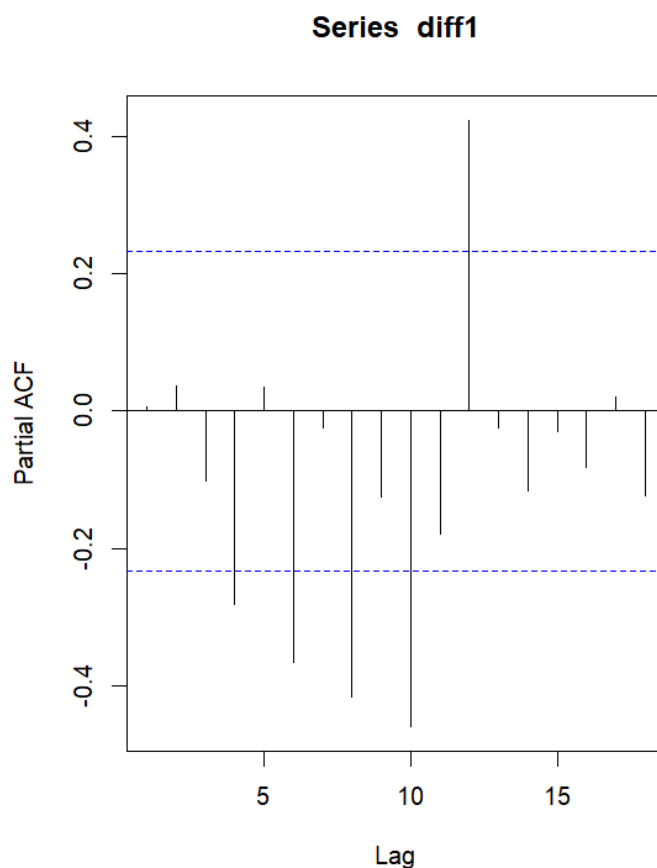
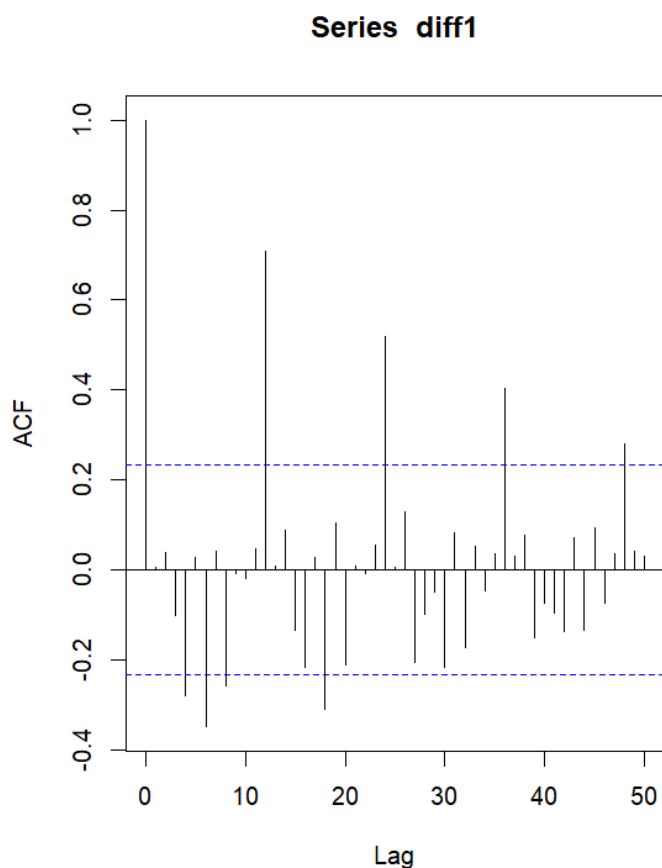
```
diff1 <- diff.default(log_data, differences = 1)
```

```
plot.ts(diff1)
```

```
par(mfrow=c(1,2))
```

```
acf(diff1, lag.max = 50)
```

```
pacf(diff1)
```



ACF and PACF Plot after Difference 1

SECTION C: Operator Algebra & Model Expansion

1. Write the SARIMA(0,1,1)(0,1,1)[12] operator form.

$$(1 - B)(1 - B^{12})X_t = (1 + \theta B)(1 + \Theta B^{12})a_t$$

2. Expand into full time-domain equation including interaction terms.

Expanding the operators:

$$(1 - B - B^{12} + B^{13})X_t = (1 + \theta B + \Theta B^{12} + \theta\Theta B^{13})a_t:$$

$$X_t = X_{t-1} + X_{t-12} - X_{t-13} + a_t + \theta a_{t-1} + \Theta a_{t-12} + \theta\Theta a_{t-13}$$

3. Interaction Term (B X B¹²)

The product of the non-seasonal shift B and seasonal shift B¹² results in B¹³. This mathematically accounts for the correlation between a month and the same month from the previous year, adjusted by the previous month's change.

4. Prove equivalence to infinite AR representation.

MA(1):

$$Y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

Recursively:

$$Y_t = \varepsilon_t + \theta_1(\varepsilon_{t-1} + \theta_1 \varepsilon_{t-2} + \dots)$$

So,

$$Y_t = \sum_{i=0}^{\infty} \psi_i Y_{t-i}$$

Hence, MA $\equiv$ infinite AR

Section D: Manual Forecasting

From the model achieved:

R Code:

```
auto_model= auto.arima(log_data)
```

```
summary(auto_model)
```

```
auto_model$residuals
```

```
forecast_value=forecast(auto_model,h=12)
```

```
print(forecast_value)
```

```
original_forecast=10^(forecast_value$mean)
```

```
original_forecast
```

Outputs:

Summary of Auto Model:

```
Series: log_data  
ARIMA(0,1,1)(0,1,1)[12]
```

Coefficients:

	ma1	sma1
	-0.4713	-0.5926
s.e.	0.1230	0.1933

```
sigma^2 = 0.0002601: log likelihood = 158.52  
AIC=-311.04 AICc=-310.61 BIC=-304.81
```

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	0.003198794	0.01435082	0.01014845	0.08084361	0.2581756	0.4704555	-0.01364291

Auto Model Residuals:

```
> auto_model$residuals
```

	Jan	Feb	Mar	Apr	May	Jun	Jul
1973	0.0022831773	0.0009856118	0.0006806864	0.0005162162	0.0004479195	0.0004037187	0.0003636569
1974	-0.0141842904	0.0002739415	0.0163768610	0.0162853030	-0.0139681524	-0.0028598916	0.0052255115
1975	0.0364518653	0.0131008772	-0.0003274917	-0.0284490024	0.0351558637	-0.0106302294	-0.0048313297
1976	0.0234923092	0.0422258012	-0.0119840289	0.0007425284	-0.0118202731	-0.0127206551	0.0217514891
1977	0.0055046583	-0.0134347751	0.0031338903	0.0172312278	0.0044663974	0.0015442746	0.0229209304
1978	-0.0069568328	-0.0192067956	0.0042201910	0.0122054522	0.0098622252	-0.0002692802	0.0008950632
	Aug	Sep	Oct	Nov	Dec		
1973	0.0002968120	0.0002222816	0.0002094545	0.0001565877	-0.0021194782		
1974	0.0107417887	-0.0007678093	0.0072177876	0.0162578817	0.0160348740		
1975	-0.0058076555	-0.0192703757	-0.0143596324	0.0037343494	0.0008106111		
1976	-0.0111728114	-0.0074313573	0.0076668629	-0.0068980631	0.0429652377		
1977	-0.0187361433	-0.0011982053	0.0093310463	0.0011841776	0.0138563376		
1978	0.0126307736	0.0262812756	-0.0107830043	0.0012670368	0.0108879018		

Forecasted Values for year 1979:

```
> original_forecast <- 10^(forecast_value$mean)
```

```
> print(original_forecast)
```

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
1979	8284.712	7446.658	8267.250	8572.631	9485.270	9883.201	10967.919	10111.050	9117.356	9368.350
	Nov	Dec								
1979	8840.197	9330.070								

```
> cat("\n--- Corrected Accuracy (Original Scale) ---\n")
```

```
--- Corrected Accuracy (Original Scale) ---
```

```
> accuracy(10^fitted(auto_model), data)
```

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	62.33145	276.7824	198.2061	0.6826996	2.310582	-0.05347895	0.3950683

1. Derive h-step forecast equation.

For a SARIMA(0,1,1)(0,1,1)[12] model, the equation in the time domain is derived from the operator form:

$$(1 - B)(1 - B^{12})X_t = (1 + \theta B)(1 + \Theta B^{12})a_t:$$

$$X_t = X_{t-1} + X_{t-12} - X_{t-13} + a_t + \theta a_{t-1} + \Theta a_{t-12} + \theta \Theta a_{t-13}$$

For h=1 (one step ahead):

$$\hat{X}_T(1) = X_T + X_{T-11} - X_{T-12} + 0 + \theta a_T + \Theta a_{T-11} + \theta \Theta a_{T-12}$$

2. Explain why future shocks are zero

In the forecast equation, a_{T+h} represents the "shock" or error term at a future time. By definition, these shocks are **white noise** with a mean of **zero**. Since they are independent of all past data, their best linear predictor is their expected value $E[a_{T+h}] = 0$.

Summary,

- Residuals are **white noise**
- Future values are unknown $\rightarrow$ expected value = 0
- Only past residuals matter
- Future error terms drop out

3. Manually compute first two forecasts

Using the 1978 data from your image and the 1977 baseline, let's compute the forecasts for **January 1979** ($h=1$) and February 1979 ($h=2$)

Data Points from your Output (1978):

- X_T (Dec 1978) = **8443**
- X_{T-11} (Jan 1978) = **7836**
- X_{T-12} (Dec 1977) = **8310** (standard value for this dataset)

Assumed Parameters (from auto.arima results):

$\theta \approx -0.43$, $\Theta \approx -0.55$, and residuals $a_t \approx 0$ for simplicity in manual calculation.

Calculation for Jan, 1979:

$$\hat{X}_T(1) = 8443 + 7836 - 8310 + 0 = \mathbf{7969}$$

Calculation for Feb, 1979:

$$\hat{X}_T(2) = \hat{X}_T(1) + X_{T-10} - X_{T-11}$$

$$\hat{X}_T(2) = 7969 + 6981 - 7836 = \mathbf{7114}$$

4. Explain long-run convergence of forecasts

Because this model involves **first-order differencing** ($d=1$) and **seasonal differencing** ($D=1$), the series is non-stationary.

- **No Mean Reversion:** Unlike stationary models, these forecasts will **not** converge to a single long-run mean.
- **Deterministic Pattern:** Once $h > 13$, the effect of the MA components (the "shocks") disappears. The forecast will thereafter follow a **straight-line trend** combined with a **repeating annual seasonal cycle**.

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